

Avionics Navigation Systems

ESE482\ECE4635 - Fall 2026

Lecture 4

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Lecture Topic

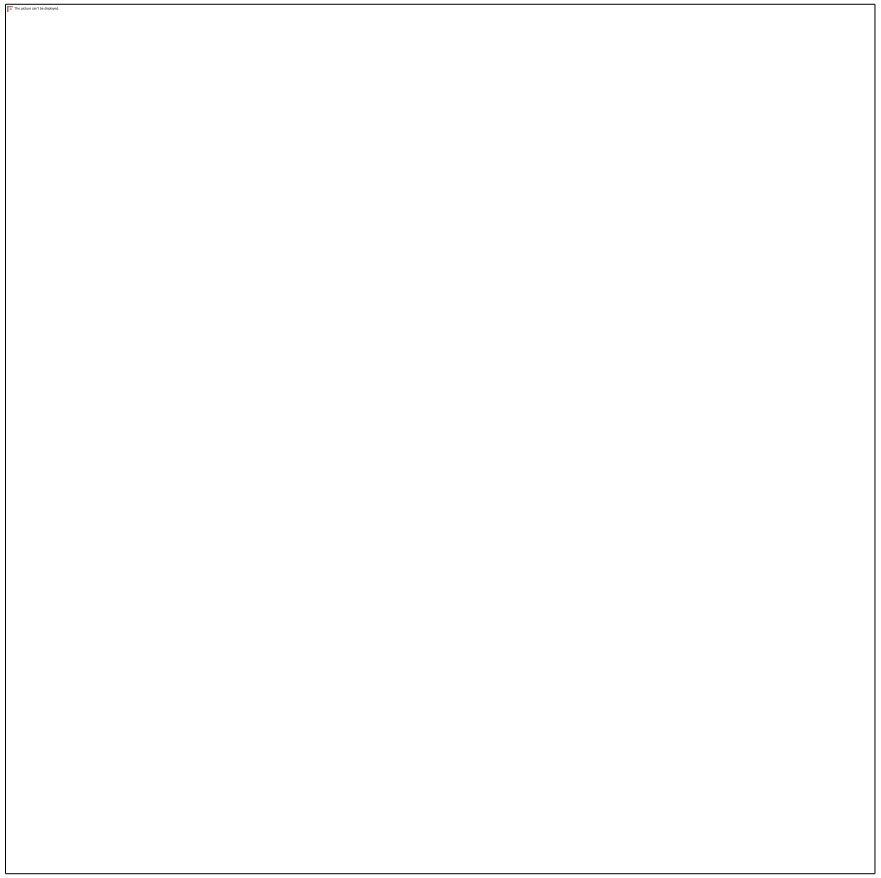
Multi-sensor Navigation Systems

Chapter # 3

Introduction

Chapter Objectives

- Explain the concept of multisensor navigation.
- Identify the limitations of individual navigation systems.
- Explain why navigation systems are integrated.
- Describe GPS/INS integration.
- Explain the basic operation of a Kalman filter.
- Distinguish between loosely coupled and tightly coupled integration.
- Explain the importance of navigation sensor error estimation.
- Describe the role of fault detection in integrated navigation.



Introduction

Introduction

➢ **Multisensor navigation** means that the navigation computer does not rely on a single sensor. Instead, it combines measurements from multiple sensors to obtain a more accurate estimate of the aircraft's state:

Position + Velocity + Attitude

The **Navigation Sensors** divided into two groups:

1. Dead-Reckoning Sensors

These sensors measure: **Acceleration, Velocity, Angular motion**

The measurements are then **integrated over time** to determine the aircraft's position.

✓ Examples include:

- INS – Inertial Navigation System
- Doppler Radar
- Air-data sensors

2. Positioning Sensors

These sensors provide a **position measurement** that is directly related to the Earth's coordinate system.

✓ For example : Loran , GPS , Star tracker

What is a Multisensor Navigation System?

A **multisensor navigation system** combines measurements from two or more navigation systems and processes them together to determine the aircraft's:

- Position
- Velocity
- Attitude
- Heading

✓ For example: **GPS + INS → Integrated Navigation System**

The computer receives measurements from both systems and produces a better navigation solution than either system could provide independently.

Why Do We Need Multisensor Navigation?

No individual navigation system is perfect.

✓ For example: **INS**

Advantages

- Self-contained
- Does not require external signals
- Provides continuous position, velocity, and attitude

Disadvantage

Errors accumulate with time. Because the **INS relies on integration**.

✓ **GPS**

Advantages

- Provides accurate position and velocity information
- Does not suffer from inertial drift

Disadvantages

- Requires satellite signals
- Can be affected by signal blockage, interference, jamming, satellite availability and multipath.

❑ Therefore: Combine complementary navigation systems one of the most common approaches is to use an **INS together with complementary sensors** to reduce the errors that accumulate over time.

Why Do We Need Multisensor Navigation?

✓ Doppler Radar

Advantages

It provides excellent information about: Ground Velocity

Disadvantage : it requires **radar measurements**.

✓ Air Data

Advantages

It provides:

- Airspeed
- Angle of attack
- Sideslip

Disadvantage : the main problem is: Wind uncertainty because **airspeed is not the same as ground speed**.

❖ Therefore, the main idea is: Use the strengths of one sensor to compensate for the weaknesses of another

This is the basic concept of **Multisensor Navigation**.

Why Do We Need Multisensor Navigation?

System	Main Advantage	Main Limitation
INS	Completely self-contained; continuous output	Errors accumulate with time
GPS	High position accuracy	Depends on satellite signals
Radio navigation	Useful external reference	Requires suitable ground/space signals
Air-data system	Provides airspeed, altitude, etc.	Does not directly provide complete position

Basic Multisensor Navigation Concept

The basic principle can be represented as:

Sensors → Measurements → Navigation Computer → Navigation Solution

✓ **For example:**

GPS measurements

↓

INS measurements → **Navigation Computer** → Position + Velocity + Attitude

↑

Air-data measurements

✓ The computer does not simply average the measurements. It uses mathematical algorithms to determine the most probable navigation state.

Aircraft Navigation State

The navigation system needs to determine several quantities.

- **Position**

Latitude, Longitude, Altitude

- **Velocity**

V_N, V_E, V_D where:

V_N = north velocity

V_E = east velocity

V_D = downward velocity

- **Attitude**

Roll, Pitch, Yaw

✓ Therefore, the navigation state can be represented generally as:

x=[Position, Velocity, Attitude, Sensor Errors]

Inertial System Characteristics

Why does an INS need aiding sensors?

The INS is mainly based on two types of sensors:

1. **Accelerometers :** They measure **specific force / acceleration**.
2. **Gyroscopes :** They measure **angular change**.

❑ INS Mechanization

➢ The accelerometer measurements are **not used directly**. The overall process is approximately:

Accelerometers → Force measurements → Transform to Navigation Frame → Gravity correction → Coriolis correction → Acceleration → Integration → Velocity → Integration → Position

➢ The gyroscopes, on the other hand, follow this process:

Gyroscopes → Angular measurements → Orientation computation → Navigation-frame orientation

✓ The accelerometer measurements are first transformed into the **navigation coordinate frame**.
✓ The effects of **gravity** and **Coriolis acceleration** are then accounted for. The resulting acceleration is integrated to obtain **velocity**, and velocity is integrated again to obtain **position**.

INS Errors

Where is the Problem?

Suppose we have a very small error in the measured acceleration:

$$a_{measured} = a_{true} + \Delta a$$

After the first integration:

$$\Delta V = \int \Delta\, a\, dt$$

Then, after integrating the velocity error:

$$\Delta P = \int \Delta\, V\, dt$$

Therefore: **Acceleration Error → Velocity Error → Position Error**

- ✓ This is one of the fundamental characteristics of an **inertial navigation system**.
- ✓ A small sensor error does not simply remain a small error. Because the INS obtains velocity and position through **integration**, the error can accumulate with time.
- ✓ Therefore, an INS **cannot rely on itself indefinitely without correction**. It needs additional measurements from **aiding sensors** to limit or correct the accumulated errors.

INS Errors

Schuler Oscillation

The errors in an inertial system used for **Earth-referenced navigation** can exhibit a characteristic oscillatory behavior known as: **Schuler Oscillation**

- The **Schuler period** is approximately: 84.4 minutes

This means that the behavior of INS navigation errors is **not simply**: **error** $\propto t^2$ in a straight forward way.
The reason is that the aircraft is navigating relative to the **Earth**, while the navigation frame itself is affected by the Earth's curvature and motion.
As a result, the interaction between: Earth curvature, gravity, the navigation frame, and inertial measurement errors causes certain navigation errors to behave **oscillatorily**, rather than simply growing without bound as a simple t^2 relationship.

- **The key idea**

Think of the INS as continuously calculating where the aircraft is based on its own measurements:
Sensor measurements → INS computation → Velocity → Position
If the measurements contain errors:
Small sensor error → Integration → Velocity error → Integration → Position error
And when this process is performed in an **Earth-referenced navigation system**, the Earth-related dynamics introduce the **Schuler oscillation**.

- ✓ So, the important takeaway is: **INS errors are strongly influenced by integration and Earth−referenced dynamics**

and this is one of the main reasons why **aiding sensors are used with INS**.

Integrated Stellar-Inertial System

Now begins to add an additional sensor to the INS.

➢ **Star Tracker :** A **star tracker** observes stars and determines their directions.

The basic idea is: Stars provide an approximately fixed reference in inertial space.

Meanwhile, the INS has a fundamental problem associated with: **Gyro Drift**

Therefore: **INS → Navigation solution → Gyro drift causes errors**

Star Tracker → Independent attitude reference → Estimate / correct gyro drift

star tracker can be used to calibrate gyro drift, which is one of the major sources of degradation in the accuracy of a **free-inertial system**.

❑ Why is Stellar + INS Useful?

➢ The **INS** provides:

- **High bandwidth**
- **Continuous navigation solution**
- **Position**
- **Velocity**
- **Attitude**

However, the INS is subject to **drift**.

➢ The **Star Tracker** provides: **An accurate celestial reference**

However, it does not necessarily provide the same **continuous, high-bandwidth navigation behavior** as the INS.

Therefore: **INS + Star Tracker** can provide a better navigation solution than relying on either system alone.

The basic principle is:

✓ **The INS provides continuous navigation, while the star tracker provides an independent reference that can be used to limit or correct inertial drift.**

Integrated Doppler-Inertial Systems

Here, we use: **INS + Doppler Radar**

➢ The **Doppler radar** provides: V_g where V_g represents **ground velocity**. It can also provide the **drift angle**: δ

From these measurements, we can obtain the **ground-velocity components**.

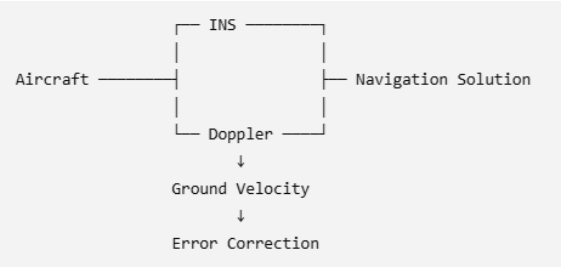
Therefore, the Doppler radar provides an: External Velocity Reference for the INS.

❖ **What Happens to the INS?**

❑ **Without Doppler :** INS → Acceleration → integration → Velocity → integration → Position → Errors grow

The INS continuously integrates its measurements. Therefore, errors in the inertial measurements can accumulate and affect the calculated velocity and position.

❑ **With Doppler**



Integrated Doppler-Inertial Systems

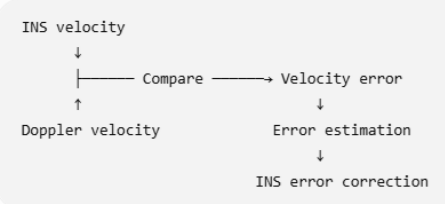
➢ The Doppler radar provides an **independent measurement of velocity**.
This velocity can be compared with the velocity calculated by the INS.
If: $\boldsymbol{V_{INS} \neq V_{Doppler}}$ then the difference is: $\boxed{\boldsymbol{\Delta V = V_{INS} - V_{Doppler}}}$

This **velocity difference** contains information about errors in the INS navigation solution.

➢ How Does This Help?

Suppose the INS calculates: $V_{INS} = 250$ m/s , while the Doppler radar measures: $V_{Doppler} = 248$ m/s
Then: $\Delta V = 250 - 248 = 2$ m/s

✓ The navigation system can use this discrepancy as an **error measurement**.
✓ Instead of allowing the INS to continue accumulating the error, the integrated system uses the external Doppler measurement to estimate and correct the inertial errors.



✓ So the Doppler radar does **not replace the INS**. Rather, it provides an independent measurement that helps the INS **correct its accumulated errors**.

➢ Important Advantage

✓ The book points out that adding a **speed sensor** to an INS can help remove or reduce **Schuler oscillations** in the position and velocity errors.
✓ It can also help correct **orientation errors**.
✓ Therefore, the main concept of an integrated Doppler-inertial system is: **INS provides continuous navigation + Doppler provides external velocity information**
✓ which allows the system to maintain a more accurate navigation solution over time.

Airspeed-Damped Inertial System

➢ Instead of using **Doppler Radar**, we can use: **Air Data**

The **air-data system** typically provides:

- **Airspeed**
- **Angle of attack** α
- **Sideslip** β

Using the **attitude information** provided by the INS, the air-relative velocity can then be transformed into an **Earth-referenced velocity**.

However, there is an important problem: Wind

Because: $\vec{V}_{ground} = \vec{V}_{air} + \vec{V}_{wind}$

Therefore, if the wind is not accurately known, the calculated **ground velocity** will also contain an error.

✓ **uncertainty in the wind** is the main source of difficulty when comparing the velocity derived from air-data measurements with the velocity calculated by the INS.

➢ Why Does Airspeed Not Directly Give Ground Velocity?

The aircraft has two different velocity concepts:

1. **Air-relative velocity :** This describes how the aircraft moves **relative to the surrounding air**. $\vec{V}_{air}$
2. **Ground velocity :** This describes how the aircraft moves **relative to the Earth**. $\vec{V}_{ground}$

The wind connects the two: $\vec{V}_{ground} = \vec{V}_{air} + \vec{V}_{wind}$

➢ For example, if an aircraft has an airspeed of $250\ m/s$, this does **not** automatically mean that its ground speed is $250\ m/s$.

A headwind or tailwind can change the ground velocity.

Therefore: Air Data → Air-relative velocity → Need wind information → Ground velocity

✓ This is why **wind uncertainty** becomes important in an Airspeed-Damped INS.

Airspeed-Damped Inertial System

➢ The key difference

The most important distinction between **Doppler-INS** and **Airspeed-INS** is the reference used for the velocity measurement:

$$\boxed{\text{Doppler} \rightarrow \text{Ground}-\text{referenced velocity}}$$

while:

$$\boxed{\text{Air Data} \rightarrow \text{Air}-\text{referenced velocity}}$$

✓ Therefore, Doppler measurements can provide a more direct external reference for ground velocity, whereas air-data measurements require knowledge or estimation of the **wind vector**.

Airspeed-Damped Inertial System

Important Comparison

System	External Reference	Main Benefit	Main Problem
INS	None	Autonomous, continuous navigation	Errors grow with time
Stellar-INS	Stars	Corrects/estimates gyro drift	Requires star visibility and tracking
Doppler-INS	Ground velocity	Helps bound velocity and navigation errors	Requires Doppler radar
Airspeed-INS	Air-relative velocity	Uses existing air-data measurements	Wind uncertainty
GPS-INS	GPS position/velocity	Provides long-term external correction	Requires external satellite signals

Integrated Stellar-Inertial-Doppler System

The next step is to combine **three complementary sources of information**: INS + Star Tracker + Doppler Radar

➢ **INS**

The **INS** provides: **Position** , **Velocity** , **Attitude** , **High bandwidth**
However, its measurements are affected by errors that accumulate over time.

➢ **Star Tracker**

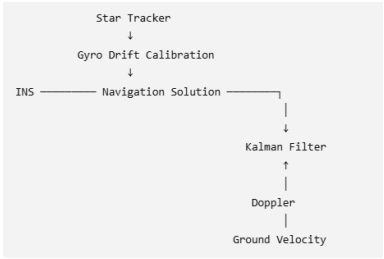
The **Star Tracker** helps with: Gyro Drift Calibration
It provides an accurate **celestial/inertial attitude reference**, which can be used to estimate and calibrate gyro drift.

➢ **Doppler Radar**

The **Doppler Radar** provides: Velocity Reference
More specifically, it provides an independent measurement of **ground velocity**.

❑ **Integrated System**

The three systems can therefore work together:



➢ **How does the system work?**

The **INS** continuously produces the navigation solution: Position + Velocity + Attitude

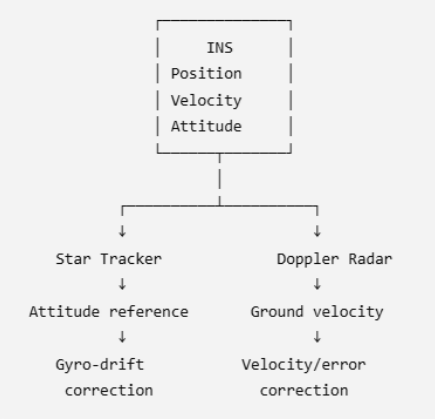
✓ Because the INS operates at a **high bandwidth**, it can provide continuous navigation information.
✓ At the same time, the **Star Tracker** provides an independent attitude reference. This information can be used to estimate and calibrate **gyro drift**, which is one of the important sources of inertial navigation error.
✓ The **Doppler Radar**, meanwhile, provides an independent measurement of **ground velocity**.
✓ The system can then compare the velocity calculated by the INS with the velocity measured by the Doppler radar: $\Delta V = V_{INS} - V_{Doppler}$
✓ These differences provide information about the errors in the inertial solution.

Integrated Stellar-Inertial-Doppler System

- The idea is to use each sensor to address a different weakness of the inertial system:

INS + Star Tracker + Doppler Radar

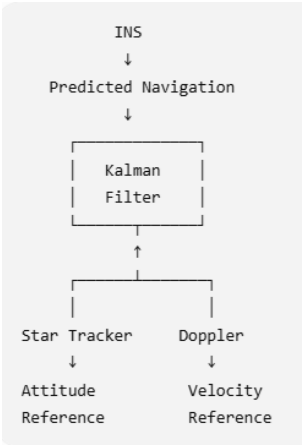
- ✓ So the overall concept is: INS + Independent Attitude Reference + Independent Velocity Reference
- ✓ The **Star Tracker** primarily helps with the **attitude/gyro-drift problem**, while the **Doppler Radar** provides an independent **velocity reference**.
- ✓ This is a good example of the central idea of **multisensor navigation**:
- ✓ **Different sensors provide independent information that can be used to estimate and correct different INS error sources.**



Integrated Stellar-Inertial-Doppler System

➢ Why is the Kalman Filter Important?

The **Kalman filter** acts as the mathematical mechanism that combines these different measurements.



✓ So the Kalman filter does not simply choose one sensor over another. It uses the **different measurements and their associated errors/uncertainties** to estimate the navigation state and the INS error state.

➢ **The key idea :** The roles of the sensors can be summarized as:

INS	→	Continuous navigation solution
Star Tracker	→	Attitude / gyro−drift reference
Doppler	→	Ground−velocity reference
Kalman Filter	→	Sensor fusion and error estimation

✓ The **stellar measurements can help calibrate gyro drift**, after which the velocities produced by the stellar-inertial system can be compared with measurements from a velocity sensor such as Doppler. This combination provides additional information that can be used for improved **calibration and error estimation**.

✓ **In one sentence: The INS provides the continuous navigation solution, the Star Tracker helps control attitude/gyro-drift errors, the Doppler provides an independent velocity reference, and the Kalman filter combines all of this information to estimate and correct the navigation errors.**

Position Update of an Inertial System

One of the simplest ways to reduce **INS position drift** is to periodically provide the INS with a **known or more accurate position reference**.

- For example, the INS provides:

$$\phi_{INS}, \lambda_{INS}$$

where $\phi$is latitude and $\lambda$is longitude.

- Suppose we also have GPS measurements:

$$\phi_{GPS}, \lambda_{GPS}$$

We can then calculate the difference between the INS position and the GPS position:

$$\boxed{\Delta P = P_{INS} - P_{GPS}}$$

✓ This difference is commonly referred to as the **position error** or **measurement residual**.

Position Update of an Inertial System

➢ How Does the Position Update Help?

Consider the basic INS operation:

Accelerometers → Integration → Velocity → Integration → Position → Errors accumulate

✓ Without an external position reference, the INS continues integrating its measurements, so small errors can gradually produce a significant position error.

With GPS updates: INS → INS Position → Compare → GPS Position → Position Difference → Kalman Filter → Error Estimation → INS Correction

✓ The GPS does **not necessarily replace the INS position solution**. Instead, it provides an **external reference** that allows the system to determine how much the INS solution has drifted.

Position Update of an Inertial System

➢ **Position Update Can Correct More Than Position**

This is one of the most important ideas in an **integrated navigation system**.
At first, we might think: GPS gives position → therefore we can only correct position.

But the **Kalman filter** can use the position residual to estimate other INS errors as well.

❑ **For example:** Position Difference → Estimate INS Errors

These can include:

- **Position errors**
- **Velocity errors**
- **Orientation errors**
- **Sensor errors**

✓ So the **position measurement acts as an indirect source of information about other errors in the inertial system**.

Position Update of an Inertial System

❑ Example

Suppose: $P_{INS} = P_{GPS} + 100$ m
Then: $\Delta P = P_{INS} - P_{GPS} = 100$ m

The system knows that the INS solution has deviated from the external position reference.

➢ The Kalman filter can then determine whether this position discrepancy is consistent with:
- accumulated velocity error,
- attitude error,
- accelerometer bias,
- gyro errors,
- or other INS error sources.

It then estimates the corresponding error states and applies appropriate corrections.

Position Update of an Inertial System

➢ What Can Provide the Position Update?

position updates can come from external navigation systems such as:

- **Loran**
- **Omega**
- **GPS**

GPS is particularly important in modern integrated navigation systems because it can provide highly useful external position and velocity measurements.

➢ Therefore, the overall principle is:

INS + Periodic Position Updates → Reduced Navigation Drift

➢ And with Kalman filtering:

Position Measurement → Error Estimation → INS Correction

✓ **The key concept to remember :**

A position update does more than correct position. Through the Kalman filter, the position residual can provide information about other INS error states, including velocity, orientation, and sensor errors.

Noninertial GPS Multisensor Navigation Systems

- Not every **multisensor navigation system** has to be built around an INS.
- **For example,** a multisensor system could combine: GPS + Doppler or: GPS + Loran

The examples of **GPS receiver modules integrated into systems such as Doppler radar or Loran**.

- The important idea is that **multisensor navigation simply means combining information from multiple navigation sensors**. The system architecture does not necessarily have to include an INS.
- However, the **INS remains very important** because it provides a:
  - Continuous navigation solution
  - High-bandwidth reference
  - Position
  - Velocity
  - Attitude
- External sensors such as GPS, Doppler, or Loran can then provide **periodic or independent measurements** that help correct the errors of the navigation solution.

✓ **The key idea :** Multisensor Navigation ≠ INS + Sensors only

Rather: Multisensor Navigation = Combining complementary navigation measurements

✓ The reason INS-based systems are so common is that the INS can provide the **continuous, high-bandwidth navigation reference**, while other sensors provide external information to **bound or correct the accumulated inertial errors**.

Filtering of Measurements

- In a multisensor navigation system, we have several measurements, and **each measurement contains some level of noise or error**.

❑ For example: $GPS = P_{true} + noise$

while the INS can be represented conceptually as: $\boldsymbol{INS = P_{true} + drift}$

The objective is to combine the available measurements and obtain a **better estimate of the true state**.

➢ A Simple Example

Suppose we have: **GPS** $100.0\ m$, **INS** $102.0\ m$

We cannot simply take the average:

$$P = \frac{100 + 102}{2} = 101\ m$$

because the two sensors do **not necessarily have the same accuracy or uncertainty**.

- For example
- ✓ if GPS has a smaller measurement error, we should give the GPS measurement **more weight**.
- ✓ If the INS estimate is more reliable at a particular instant, then the INS should receive **more weight**.
- ✓ This leads to the fundamental concept of: Weighting

Filtering of Measurements

➢ Why Do We Need Weighting?

Imagine that we have two measurements:
GPS → 100 m, Fusion INS → 102 m

The navigation computer needs to decide: **How much should I trust each measurement?**
This depends on the uncertainty associated with each sensor.

➢ For example:

GPS uncertainty = small error
INS uncertainty = larger error
Then the estimated position should be closer to the GPS measurement.

- ✓ Conversely, if GPS becomes less reliable while the INS estimate is relatively accurate, the system should rely more heavily on the INS.
- ✓ So the weighting is **not fixed**.

Filtering of Measurements

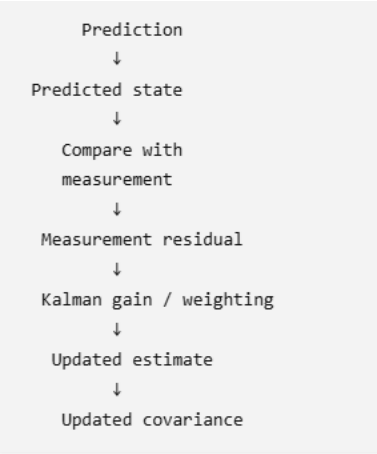
➢ Kalman Filter

This is where the **Kalman Filter** becomes important.

- The Kalman filter provides a mathematical framework for combining:
  - **The predicted state** from the navigation system
  - **New measurements** from external sensors
  - **The estimated uncertainty** associated with both

- The filter essentially asks: **How much should I trust my prediction, and how much should I trust the new measurement?**

The answer depends on the estimated errors and their **covariances**.



➢ Covariance Is Important

The **covariance** represents information about the uncertainty of the estimated states and the relationships between their errors.
Very roughly:

Large uncertainty → Less confidence

and:

Small uncertainty → More confidence

Therefore, the Kalman filter dynamically adjusts how much influence each measurement has on the final estimate.

The Basic Kalman Filter Idea

- The process can be summarized as:

Prediction → Predicted state → Compare with measurement → Measurement residual → Kalman gain / weighting → Updated estimate → Updated covariance

The difference between the predicted measurement and the actual measurement is called the **measurement residual** or **innovation**.

$$\boxed{\text{Residual} = \text{Measurement} - \text{Prediction}}$$

The filter uses this residual together with the estimated uncertainties to determine how much the state estimate should be corrected.

The Basic Kalman Filter Idea

➢ Connecting This to INS + GPS

This is exactly what we discussed in the **Position Update** section.

- The INS predicts the aircraft's position: P_{INS}
- GPS provides an external measurement: P_{GPS}

The difference is: $\boldsymbol{\Delta P = P_{INS} - P_{GPS}}$

✓ The Kalman filter uses this information to estimate the underlying INS errors and update the navigation solution.

➢ The key concept

The Kalman filter does not simply average sensor measurements. It combines measurements according to their estimated uncertainties and continuously updates the weighting as the system evolves.

Therefore: **Kalman Filtering = Prediction + Measurement Update + Uncertainty–based Weighting**

✓ This is why Kalman filtering is so central to **multisensor navigation**: it provides a systematic way to combine sensors that have **different error characteristics** and extract a better estimate of the aircraft's true navigation state.

Kalman Filter Basics

➢ **What is a Kalman Filter?**

A **Kalman Filter** is an algorithm that combines:

1. **Model prediction**
2. **Sensor measurements**

to obtain: **Best Estimate of the System State**

➢ In navigation, the **state** may include quantities such as:

- Position
- Velocity
- Attitude
- Sensor biases
- Other navigation errors

➢ **Basic Idea**

The Kalman filter works **recursively**. It does not need to start the calculation from scratch every time a new measurement arrives.

The basic process is:

Previous Estimate → Prediction → Predicted State → Compare with Measurement → Measurement Residual → Kalman Gain → Corrected Estimate → New Estimate → Repeat

✓ So the filter continuously moves between two steps:

Prediction → Measurement Update

Kalman Filter Basics

1. Prediction

First, the filter uses the **system model** and the previous estimate to predict the current state.

❑ For example, in an INS: **Previous Position + Previous Velocity + Motion Model → Predicted Position**

The prediction represents:

Where do we expect the aircraft to be based on what we already know?

2. Measurement

A sensor then provides a new measurement.

❑ For example: P_{GPS}

The filter compares this measurement with its prediction.

3. Measurement Residual

The difference between the measurement and the predicted measurement is called the **measurement residual** or **innovation**.

Conceptually: $\boxed{r = z - \hat{z}}$

where:

- z= actual sensor measurement
- $\hat{z}$= predicted measurement
- r= measurement residual

❑ For example, if: $P_{predicted} = 102\,m$ and $P_{GPS} = 100\,m$

then: $r = 100 - 102 = -2\,m$

✓ The residual tells the filter: **There is a discrepancy between what I predicted and what the sensor measured.**

Kalman Filter Basics

4. Kalman Gain

The Kalman gain determines **how strongly the new measurement should influence the current estimate**.

Residual → Kalman Gain → Size of Correction

- If the sensor measurement is considered very reliable, the filter gives it **more influence**.
- If the sensor measurement is highly uncertain, the filter gives it **less influence**.
- Similarly, if the prediction is very uncertain, the filter can place more weight on the new measurement.
- Therefore, the Kalman gain depends on the relative uncertainties of:

Prediction vs.Measurement

5. Corrected Estimate

The filter then uses the residual and Kalman gain to correct the predicted state.

$$\textbf{Corrected Estimate} = \textbf{Predicted Estimate} + \textbf{Correction}$$

and the correction is related to:

$$\textit{Correction} = K \times \textit{Residual}$$

where K is the **Kalman gain**.

$$\hat{x}_{updated} = \hat{x}_{predicted} + K(z - \hat{z})$$

This equation captures the basic intuition behind the Kalman filter.

Simple Navigation Example

- Suppose the INS predicts: $P_{INS} = 102\,m$
- GPS measures: $P_{GPS} = 100\,m$

Therefore:

$$r = P_{GPS} - P_{INS}$$
$$r = 100 - 102 = -2\,m$$

Now suppose the Kalman filter determines that the appropriate gain is:

$$K = 0.75$$

Then the correction is:

$$Kr = 0.75(-2) = -1.5\,m$$

Therefore:

$$P_{updated} = 102 - 1.5$$
$$\boxed{P_{updated} = 100.5\,m}$$

✓ Notice that the result is **not simply the average** of 102 m and 100 m.

✓ The filter has moved the estimate toward GPS according to the **relative uncertainties**.

Simple Navigation Example

Then We Repeat

After obtaining the corrected estimate, it becomes the starting point for the next cycle:

Previous Estimate → Prediction → Predicted State → New Sensor Measurement → Residual → Kalman Gain → Correction → Updated State → Repeat

- ✓ This is why the Kalman filter is called a **recursive estimator**.
- ✓ It continuously updates the navigation solution as new measurements become available.

➢ **The Most Important Idea to Remember**

Do not think of the Kalman filter as simply: **"Taking the average of two sensors."**

Instead, think of it as:

Predict → Measure → Compare → Weight → Correct → Repeat

And in an integrated navigation system:

INS prediction + External sensor measurement → Kalman Filter → Improved navigation estimate

This is the fundamental mechanism behind **INS/GPS, INS/Doppler, and other multisensor navigation systems**.

Participation

Kalman filter correction

An INS predicts an aircraft position of 102 m, while GPS measures 100 m. The Kalman filter uses a gain of $K = 0.75$.

Use the simplified scalar update:

$$\hat{x}^+ = \hat{x}^- + K(z - \hat{x}^-)$$

Calculate:

a. The measurement residual.

b. The correction applied to the predicted estimate.

c. The updated position estimate.

Participation

The residual is the difference between the GPS measurement and the INS predicted position:

$$r = z - \hat{x}^-$$

where $z = 100\,\text{m}$ and $\hat{x}^- = 102\,\text{m}$.

$$r = 100 - 102 = -2\,\text{m}$$

The correction is calculated using: $\Delta\hat{x} = Kr$

$$\Delta\hat{x} = 0.75(-2) = -1.5\,\text{m}$$

The updated estimate is:

$$\hat{x}^+ = \hat{x}^- + \Delta\hat{x}$$

$$\hat{x}^+ = 102 + (-1.5)$$

$$\boxed{\hat{x}^+ = 100.5\,\text{m}}$$

Participation

1. An aircraft is flying over an area where GPS signals become unavailable for 5 minutes. Its INS continues operating normally. Tasks

A. Explain how the aircraft can continue navigating.

B. Describe what happens to the navigation error during the GPS outage.

C. Suggest a sensor that could provide an independent velocity reference.

answer

The INS continues estimating position, velocity, and attitude using inertial measurements. Its errors can accumulate during the outage. Doppler radar can provide an independent ground-velocity reference and help limit navigation errors.

2. An aircraft's air-data system measures airspeed accurately, but the wind velocity is unknown. Tasks

A. Can airspeed alone determine the aircraft's ground velocity?

B. Explain the role of wind information.

C. Which integrated navigation approach discussed in the lecture is affected by this problem?

answer

No. Ground velocity is the vector sum of air-relative velocity and wind velocity. Without wind information, ground velocity cannot be determined accurately from airspeed alone. This is the main difficulty with Airspeed-Damped INS.

The Process and Measurement Models

➢ Process Model

We describe the **system dynamics** using: $\boldsymbol{x_{k+1} = \Phi_k x_k + w_k}$

- x_k= **state vector**
- Φ_k= **state transition matrix**
- w_k= **process noise**

The physical meaning is: **Given the current state and the system model, where do we expect the system to be at the next time step?**

A. State Vector x_k:

It is a vector that combines the variables we want to estimate using the Kalman Filter.

For example, if we want to estimate position and velocity errors along one direction:

$$\mathbf{x}_k = \begin{bmatrix} \delta p_k \\ \delta v_k \end{bmatrix}$$

- δp_k: Position error in meters.
- δv_k: Velocity error in meters per second.

✓ In an integrated GPS/INS model, the state vector may also include attitude errors, gyroscope biases, and accelerometer biases, depending on the model being used.

✓ Important: In this section, the state vector generally represents the navigation errors to be estimated, rather than necessarily representing the aircraft's actual position and velocity directly.

The Process and Measurement Models

- **Process Model**

$$\boldsymbol{x_{k+1} = \Phi_k x_k + w_k}$$

B. State Transition Matrix Φ_k

This matrix determines how the current state errors affect the state errors at the next time step.

For example, if the velocity error remains constant over a time interval Δt, the new position error is affected by the velocity error:

$$\delta p_{k+1} = \delta p_k + \Delta t\, \delta v_k$$

If we assume that the velocity error itself does not change during this interval, the model becomes:

$$\begin{bmatrix} \delta p_{k+1} \\ \delta v_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \delta p_k \\ \delta v_k \end{bmatrix}$$

$$\boldsymbol{\Phi}_k = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$$

This is a simplified example for illustration purposes, not the complete GPS/INS transition matrix.

Process Model

➢ **For example**

Suppose the current position error is $5\,m$, the velocity error is $0.2\,m/s$, and the time interval is $\Delta t = 10\,s$.

Neglect process noise in this example.

Required: Calculate the predicted position and velocity errors after 10 seconds.

$$\begin{bmatrix} \delta p_{k+1} \\ \delta v_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \delta p_k \\ \delta v_k \end{bmatrix}$$

Solution:

The current state vector is: $\mathbf{x}_k = \begin{bmatrix} 5 \\ 0.2 \end{bmatrix}$, $\mathbf{\Phi}_k = \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$

$$\mathbf{x}_{k+1} = \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 0.2 \end{bmatrix}$$

$$\mathbf{x}_{k+1} = \begin{bmatrix} 5 + 10(0.2) \\ 0.2 \end{bmatrix} = \begin{bmatrix} 7 \\ 0.2 \end{bmatrix}$$

Therefore, the predicted position error becomes $7\,m$, while the velocity error remains $0.2\,m/s$.

The main idea is that velocity error accumulates in the position error over time.

Process Model

C. Process Noise w_k

In reality, the mathematical model cannot predict everything with complete accuracy. Unexpected changes may occur in the aircraft's motion, or additional errors may arise from the sensors.

For this reason, we introduce a random term called Process Noise:

$$\boldsymbol{x_{k+1} = \Phi_k x_k + w_k}$$

We usually assume that the mean of the process noise is zero:

$$E[\mathbf{w}_k] = \mathbf{0}$$

We also assume that its covariance matrix is known:

$$E[\mathbf{w}_k \mathbf{w}_k^T] = \mathbf{Q}_k$$

Where Q_k is the Process Noise Covariance Matrix.

- A small value in Q_k : This means that we assume the model predicts the motion with greater confidence.
- A large value in Q_k : This means that we allow for greater uncertainty in how the state evolves.

This is particularly important because the aircraft may actually move in a way that differs from the mathematical prediction.

The Process and Measurement Models

➢ Measurement Model

After the Process Model predicts the state of the system, we need to compare this prediction with the actual measurements obtained from the navigation sensors.

The measurement equation is:

$$\boldsymbol{z_k = H_k x_k + v_k}$$

- z_k = **measurement vector**
- H_k = **measurement matrix**
- x_k = **state vector**
- v_k = **measurement noise**

The idea is: The sensor does not necessarily measure the complete state directly. Instead, H_k describes how the actual system state appears in the sensor measurement.

A. Measurement Vector z_k

It is the set of measurements available at a particular time.

In a GPS/INS system, the measurements may be obtained by comparing data from the two systems, such as the difference between the INS position and the GPS position, or from GPS measurements converted into a suitable model for estimating navigation errors.

The Process and Measurement Models

➢ Measurement Model

$$\boldsymbol{z_k = H_k x_k + v_k}$$

B. Measurement Matrix H_k

This matrix connects the measurements to the states we are trying to estimate.

Let us use the example of position and velocity errors:

$$\mathbf{x}_k = \begin{bmatrix} \delta p_k \\ \delta v_k \end{bmatrix}$$

If we have a direct measurement of the position error only, then:

$$\mathbf{H}_k = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$z_k = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \delta p_k \\ \delta v_k \end{bmatrix} + v_k$$

$$\boxed{z_k = \delta p_k + v_k}$$

- Notice that the measurement is directly related to the position error, but it does not directly measure the velocity error.
- Nevertheless, the Kalman Filter can sometimes improve the estimate of the velocity error as well, because position and velocity errors may be statistically correlated within the covariance matrix.

The Process and Measurement Models

➢ Measurement Model

$$\boldsymbol{z_k = H_k x_k + v_k}$$

C. Measurement Noise v_k

Measurements are not perfectly accurate. For example, GPS may be affected by receiver noise, clock errors, and signal propagation conditions.

We represent this error using the vector v_k , and we usually assume that:

$$E[\mathbf{v}_k] = \mathbf{0}$$

The covariance matrix is: $\boxed{E[\mathbf{v}_k \mathbf{v}_k^T] = \mathbf{R}_k}$

Where $\mathbf{R}_k$ is the **Measurement Noise Covariance Matrix**.

- If $\mathbf{R}_k$ is small, the measurements are assumed to be more accurate, so the filter places greater confidence in them.
- If $\mathbf{R}_k$ is large, the measurements are less accurate, so the filter relies more heavily on its previous prediction, depending on the other information available in the model.

The Process and Measurement Models

➢ What Does Q Mean?

$$\boxed{Q = \text{Process Noise Covariance}}$$

Q represents the uncertainty associated with the **system model/process**.

✓ If: Q is large
then: We have relatively little confidence in our model prediction.

✓ If: Q is small
then: We have more confidence in the model.

❑ **For example,** if the aircraft dynamics are difficult to model accurately, the process uncertainty may be relatively large.

➢ What Does R Mean?

$$\boxed{R = \text{Measurement Noise Covariance}}$$

Rrepresents the uncertainty in the **sensor measurement**.

✓ If: R is large
then: The measurement is noisy, so we trust it less.

✓ If: R is small
then: The measurement is more accurate, so we give it more weight.

✓ This is extremely important because the Kalman filter uses this uncertainty information when determining the **Kalman gain**.

Matrix	Meaning	Main Question
Φ	State transition matrix	How does the state evolve?
Q	Process-noise covariance	How much do I trust the model?
H	Measurement matrix	How does the sensor observe the state?
R	Measurement-noise covariance	How much do I trust the measurement?

Error Covariance Matrix

➢ What Is an Error Covariance Matrix?

- Let us assume that a navigation system estimates the aircraft's position, but the estimate is not 100% accurate.
- **For example,** the system estimates the position to be (1000,m), but the actual position could be (998,m) or (1003,m).
- Therefore, there is a difference between the actual value and the estimated value, which is called the **Estimation Error**.

$$\mathbf{e}_k = \mathbf{x}_k - \hat{\mathbf{x}}_k$$

x_k: The true state of the system.
$\check{x}_k$: The state estimated by the Kalman Filter.
e_k: The estimation error.
However, knowing the error at a single instant is not sufficient. We need to know the degree of uncertainty in the estimate and the relationship between different errors. This is where the matrix **P** becomes important.

$$\mathbf{P}_k = E[\mathbf{e}_k \mathbf{e}_k^T]$$

- ✓ This equation represents the error covariance matrix, assuming that the estimation error has a zero mean, or that the error covariance is calculated around the mean estimation error.
- ✓ In simpler terms, the matrix **P** indicates the degree of uncertainty in the navigation system's estimates.

Error Covariance Matrix

➢ **Form of the Error Covariance Matrix**

If the state vector contains two variables, position error and velocity error:

$$\mathbf{x}_k = \begin{bmatrix} \delta p_k \\ \delta v_k \end{bmatrix}$$

✓ then the covariance matrix will be a 2X2 matrix:

$$\boxed{\mathbf{P}_k = \begin{bmatrix} P_{pp} & P_{pv} \\ P_{vp} & P_{vv} \end{bmatrix}}$$

This matrix contains two types of information:

1. Variance

The elements on the main diagonal are:

$$P_{pp} = \mathrm{Var}(\delta p)$$
$$P_{vv} = \mathrm{Var}(\delta v)$$

They measure the amount of uncertainty in each variable individually.

For example, if the standard deviation of the position error is 3 m, then the variance is:

$$\sigma_p^2 = 3^2 = 9\,\mathrm{m}^2$$

✓ The larger the variance, the greater the uncertainty in the estimate.

Error Covariance Matrix

B. Covariance

The elements outside the main diagonal are:

$$P_{pv} = \mathrm{Cov}(\delta p, \delta v)$$
$$P_{vp} = \mathrm{Cov}(\delta v, \delta p)$$

They indicate whether the errors in the two variables are statistically related.

- Positive covariance: The errors tend to vary in the same direction.
- Negative covariance: The errors tend to vary in opposite directions.
- Zero covariance: There is no linear statistical relationship between the two errors. However, this does not necessarily mean that they are statistically independent.

For a standard real-valued covariance matrix:

$$\boxed{P_{pv} = P_{vp}}$$

✓ This means that the matrix P is symmetric.

✓ Important note: Variance alone does not describe the relationship between position errors and velocity errors. Therefore, covariance is also needed to provide information about the relationship between them.

Error Covariance Matrix

✓ Suppose that the navigation system estimates position errors in the North and East directions:

$$\mathbf{e} = \begin{bmatrix} e_N \\ e_E \end{bmatrix}$$

✓ Then, the covariance matrix becomes:

$$\boxed{\mathbf{P} = \begin{bmatrix} \sigma_N^2 & P_{NE} \\ P_{EN} & \sigma_E^2 \end{bmatrix}}$$

This matrix consists of two types of elements.

➢ First: Diagonal Elements

$$P_{NN} = \sigma_N^2, \qquad P_{EE} = \sigma_E^2$$

These elements represent the variance of each variable's error individually.

- If the variance is small, the estimate is more precise.
- If the variance is large, the uncertainty is greater.

➢ Second: Off-Diagonal Elements

$$P_{NE} = P_{EN}$$

These elements represent the covariance, which indicates the degree of statistical relationship between the errors of the two variables.

✓ For example, North and East position errors may be correlated due to common errors in the navigation system.

Error Covariance Matrix

- **Example of the Matrix**

$$\mathbf{P} = \begin{bmatrix} 4 & 1.2 \\ 1.2 & 9 \end{bmatrix}$$

— **North Variance** The standard deviation of the North position error is:

$$\sigma_N = \sqrt{4} = 2\,\mathrm{m}$$

— **East Variance** The standard deviation of the East position error is:

$$\sigma_E = \sqrt{9} = 3\,\mathrm{m}$$

— **Covariance**

Both values of 1.2 represent the covariance between the North and East position errors. Assuming that the position errors are measured in meters, the covariance unit is m^2.

Error Covariance Matrix

➢ **Why Is P So Important?**

The Kalman filter uses the covariance information to determine **how much confidence to place in the prediction and the measurement**.

➢ **Why Do We Need the Error Covariance Matrix in the Kalman Filter?**

The Kalman Filter does not simply calculate the predicted position of the aircraft; it also needs to know the degree of uncertainty in that estimate.

❑ **For example,** imagine that the aircraft uses an INS (Inertial Navigation System) to estimate its position, and then a new measurement arrives from GPS (Global Positioning System).

- If the INS estimate is inaccurate, the filter will give greater weight to a reliable GPS measurement.
- If the INS estimate is highly accurate and the GPS measurement is less reliable, the filter will rely more heavily on the previous estimate.

✓ The covariance matrix **P** helps the filter determine these weights, together with the measurement noise covariance matrix **R**.

Error Covariance Matrix

➢ **How Does the Error Covariance Matrix Change? Prediction and Update**

the error covariance matrix changes through two consecutive processes.

➢ **First Stage: Prediction**

Before a new measurement arrives, the filter uses the motion model to predict the next state and its uncertainty.

The covariance prediction equation is:

$$\mathbf{P}_k^- = \mathbf{\Phi}_{k-1}\mathbf{P}_{k-1}^+\mathbf{\Phi}_{k-1}^T + \mathbf{Q}_{k-1}$$

- $\mathbf{P}_{k-1}^+$: The covariance matrix after the previous update.
- $\mathbf{P}_k^-$: The predicted covariance matrix before the new measurement arrives.
- $\mathbf{\Phi}_{k-1}$: The **State Transition Matrix**, which describes how the state errors evolve over time.
- $\mathbf{Q}_{k-1}$: The **Process Noise Covariance Matrix**, which represents uncertainty arising from the motion model and noise affecting the system.

Why do we add Q ?

Because the motion model is not perfect. Errors may arise in accelerometers and gyroscopes, or unexpected changes may occur in the aircraft's motion.

Therefore, under typical conditions, uncertainty tends to increase as time passes without correction.

Error Covariance Matrix

➢ **Second Stage: Measurement Update**

When a new measurement arrives, such as a GPS measurement, the filter uses it to correct the estimated state and update the covariance matrix.

This information directly affects the Kalman gain.

➢ First, we calculate the Kalman Gain:

$$K_k = P_k^- H^T (H P_k^- H^T + R)^{-1}$$

- $\mathbf{K}_k$: The Kalman Gain, which determines how much influence the new measurement has on the state estimate.
- $\mathbf{H}_k$: The Measurement Matrix, which relates the estimated state to the measurements.
- $\mathbf{R}_k$: The Measurement Noise Covariance Matrix, which describes the uncertainty in the measurement.

✓ Next, the covariance matrix can be updated using the simplified formula

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)\mathbf{P}_k^-$$

✓ Where I is the Identity Matrix.

✓ The superscript +ve indicates the value after the measurement update, while the superscript -ve indicates the value before the measurement update.

✓ Important result: When a measurement provides useful and reliable information, the update generally reduces the uncertainty in the estimated state.

Error Covariance Matrix

Example :

Suppose that a navigation system estimates the aircraft's position error in one dimension only, and the following values are given:

- The predicted variance before receiving GPS is $P^- = 16\,\mathrm{m}^2$.
- The GPS measurement variance is $R = 4\,\mathrm{m}^2$.
- The measurement directly observes the position, so $H = 1$.

Required: Calculate the Kalman Gain and the variance after the measurement update.

Step 1: Calculate the Kalman Gain

For a single-variable case:

$$K = \frac{P^-}{P^- + R}$$

$$K = \frac{16}{16 + 4} = \boxed{0.8}$$

This means that the new measurement has a strong influence on the correction because the uncertainty in the previous estimate is greater than the uncertainty in the GPS measurement.

Step 2: Calculate the Variance After the Update

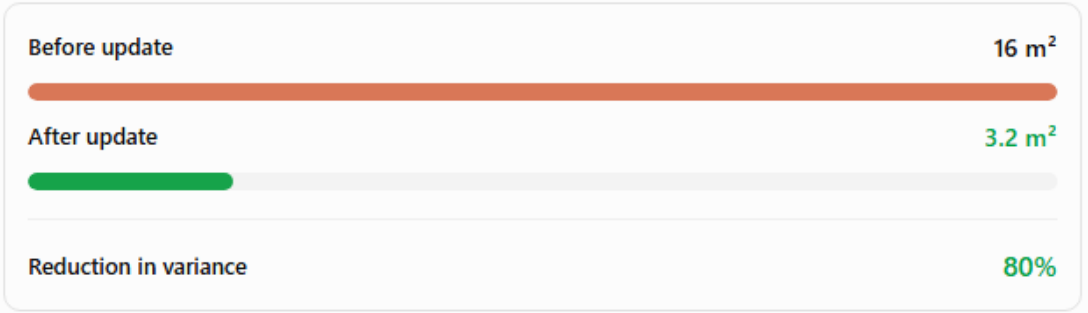
$$P^+ = (1 - K)P^-$$

Substituting $K = 0.8$ and $P^- = 16$:

$$P^+ = (1 - 0.8)(16)$$

$$\boxed{P^+ = 3.2\,\mathrm{m}^2}$$

Comparison of Uncertainty Before and After the Measurement



The variance decreased from 16 to $3.2\,\mathrm{m}^2$, representing an **80% reduction in variance.**

Therefore, the reliable GPS measurement helped the Kalman Filter reduce the uncertainty in the estimated position.

Recursive Kalman Filter

- The Kalman filter operates **recursively**.

This means that it does not need to store all previous measurements.
Instead, the current estimate and covariance contain the information needed to process the next measurement.

- The complete cycle is:

Previous Estimate → Prediction → Predicted State → Covariance Prediction → New Measurement → Measurement Residual → Kalman Gain → State Correction → Covariance Update → Next Prediction → Repeat

Recursive Kalman Filter

➢ **Step 1 – Prediction**

The predicted state is: $\boxed{\hat{x}_k^- = \Phi_{k-1}\hat{x}_{k-1}}$

The superscript **−** means: **Before incorporating the new measurement.**

So this is the filter's prediction based on the previous state and the system model.

➢ **Step 2 – Covariance Prediction**

The predicted error covariance is: $\boxed{P_k^- = \Phi P_{k-1}\Phi^T + Q}$

This equation tells us how the uncertainty changes during the prediction step.

Notice the Q: $+Q$

This makes sense because the process itself introduces uncertainty.

So: $\boxed{\text{Prediction uncertainty} = \text{Previous uncertainty propagated} + \text{Process uncertainty}}$

➢ **Step 3 – Calculate the Kalman Gain**

The Kalman gain is: $\boxed{K_k = P_k^- H^T (H P_k^- H^T + R)^{-1}}$

This is one of the most important equations in the Kalman filter.

The Kalman gain determines **how strongly the measurement should affect the predicted state**.

✓ Conceptually:

High measurement uncertainty R → Less trust in measurement → Smaller measurement influence

✓ while:

Low measurement uncertainty R → More trust in measurement → Greater measurement influence

✓ The prediction uncertainty P_k^- also plays an important role.

Recursive Kalman Filter

➢ **Step 4 – Measurement Residual**

The residual is: $\boxed{r_k = z_k - H\hat{x}_k^-}$

It represents the difference between:

- what the sensor **actually measured**, and what the filter **expected the sensor to measure** based on its predicted state.

So: $\boxed{\text{Residual} = \text{Actual Measurement} - \text{Predicted Measurement}}$

This is the information that tells the filter: **Something about my prediction does not agree with the new measurement.**

➢ **Step 5 – Correct the State**

The updated state is: $\boxed{\hat{x}_k = \hat{x}_k^- + K_k r_k}$

This equation has a very intuitive meaning: $\boxed{\text{Updated Estimate} = \text{Prediction} + \text{Weighted Correction}}$

because: $K_k r_k$

is the correction based on the measurement residual and the Kalman gain.

➢ **Step 6 – Update the Covariance**

Finally: $\boxed{P_k = (I - K_k H)P_k^-}$

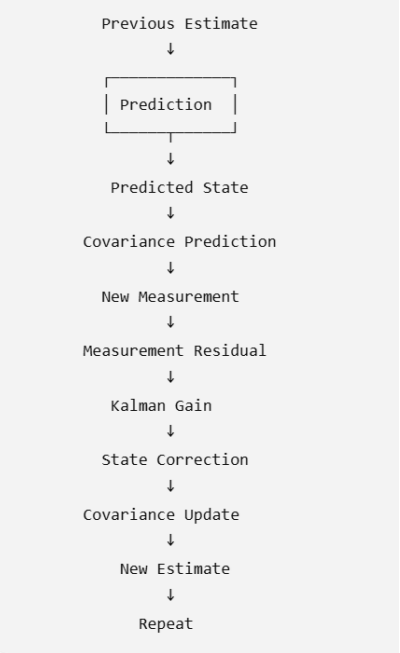
The filter updates its uncertainty after incorporating the measurement.

✓ Then the updated: $\hat{x}_k$ and: P_k become the starting point for the **next cycle**.

Recursive Kalman Filter

❑ **Putting Everything Together**

The complete recursive process is:



Or mathematically:
Predict→Measure→Compare→Weight→Correct→Repeat

The entire Kalman filter can be understood through one central idea:

$$\text{Prediction} + \text{Measurement} \rightarrow \text{Improved Estimate}$$

More precisely:

$$\text{Model Prediction} + \text{Sensor Measurement} + \text{Uncertainty Information} \rightarrow \text{Updated State Estimate}$$

And the uncertainty information comes mainly from: $Q,\ R,\ P$

where:

- Q→ **How uncertain is the process/model?**
- R→ **How uncertain is the measurement?**
- P→ **How uncertain is our current state estimate?**

Open-Loop Kalman Filter Mechanization

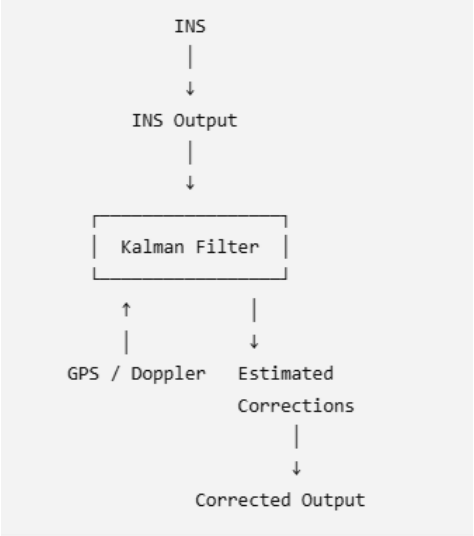
- In an **open-loop configuration**, the Kalman filter is used to correct the **outputs of the INS**.

However, the estimated corrections are **not fed back into the INS itself**.
The INS itself continues to operate **without internal correction**.

- In other words, the Kalman filter receives the INS output together with measurements from external sensors, such as GPS or Doppler Radar, and uses this information to estimate the navigation errors.

- In an **open-loop Kalman filter configuration**, the Kalman filter is used to **estimate the errors in the INS and correct its outputs**, but the corrections are **not fed back into the INS itself**.

✓ **Basic idea :** The system can be represented as:



✓ The important point is that the **INS continues operating independently**.

Open-Loop Kalman Filter Mechanization

1. What happens inside the system?

The INS continuously calculates the aircraft's:

- **Position**
- **Velocity**
- **Attitude**

However, an INS is subject to different error sources, such as:

- Gyroscope errors
- Accelerometer errors
- Initial alignment errors
- Navigation computation errors
- Sensor biases

Therefore, the INS output gradually becomes inaccurate.

✓ The Kalman filter uses additional measurements, such as **GPS or Doppler measurements**, to estimate how much error exists in the INS solution.

Open-Loop Kalman Filter Mechanization

2. What does the Kalman filter actually do?

- **For example:** Suppose the INS says: Aircraft position = 100.0 km, while GPS indicates: Aircraft position = 99.8 km

The Kalman filter processes these measurements and estimates the INS position error. Estimated INS error ≈ 0.2 km

The system can then produce: Corrected position ≈ 99.8 km

✓ But the crucial point is: **The correction is applied to the output, not to the INS internal navigation solution.**

3. Why is it called "Open-Loop"?

The term **open-loop** refers to the absence of a feedback path to the INS.

The information flows essentially in one direction: **INS → INS Output → Kalman Filter → Corrected Output**

✓ The correction does **not** return to the INS

✓ Therefore, the INS itself continues running with its original internal states.

Open-Loop Kalman Filter Mechanization

4. An important distinction Open and Closed Loop

This is the most important concept to understand:

Open-loop : The Kalman filter corrects the INS output.

The INS itself is not corrected internally.

Closed-loop : The Kalman filter corrections are fed back into the INS.

The INS internal navigation states or estimated sensor errors can therefore be corrected.

A simple comparison is:

Open-Loop	Closed-Loop
Corrects INS output	Corrects INS internally
No feedback to INS	Feedback exists
INS continues independently	INS receives corrections
Simpler architecture	More integrated architecture

Open-Loop Kalman Filter Mechanization

5. Why use an Open-Loop configuration?

One advantage is that the **INS navigation system can remain relatively independent** of the Kalman filter. The Kalman filter acts more like an additional correction layer:

INS → Navigation Solution → Kalman Filter → GPS/Doppler → Corrected Solution

The INS continues generating its normal navigation solution, while the Kalman filter estimates the errors and produces a better final output.

6. The key idea to remember

Think of the open-loop system as:

"The Kalman filter watches the INS and corrects what comes out of it, but it does not go inside the INS to fix the INS itself."

- ✓ **Open-loop Kalman filter mechanization:**
  A configuration in which the Kalman filter estimates navigation errors and applies corrections to the **INS outputs**, without feeding those corrections back into the **INS internal mechanization**.

Closed-Loop Kalman Filter

- In a **closed-loop Kalman filter configuration**, the Kalman filter does not simply correct the INS output.

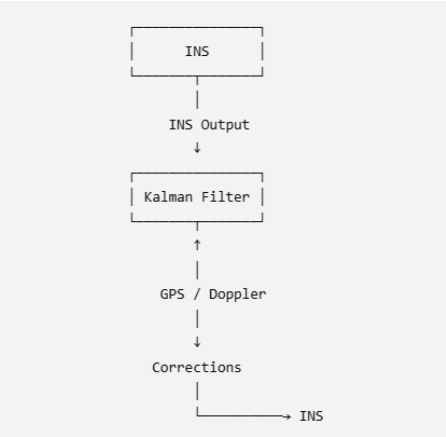
Instead:

The estimated corrections are fed back into the INS itself.

This is the main difference between **closed-loop** and **open-loop** mechanization.

1. Basic concept

The system can be represented as:



The important feature is the **feedback path** from the Kalman filter back to the INS.

Closed-Loop Kalman Filter

2. What happens in the closed-loop system?

The INS continuously produces a navigation solution:

- Position
- Velocity
- Attitude

At the same time, an external sensor such as **GPS** provides measurements that can be compared with the INS solution.
The Kalman filter processes these measurements and **estimates the errors in the INS**.

❑ **For example:** INS position = 100.0 km , GPS position = 99.8 km ,
The Kalman filter may estimate: Position error = 0.2 km

But unlike the open-loop case, the system does not simply use this correction to modify the final displayed output.
Instead, the correction is **fed back to the INS**.
Kalman Filter >> estimated error >> INS >> Updated navigation solution
Therefore, the INS itself is continuously corrected.

Closed-Loop Kalman Filter

3. What exactly can be corrected?

The Kalman filter can estimate different types of INS errors, such as:

- Position errors
- Velocity errors
- Attitude errors
- Accelerometer biases
- Gyroscope biases
- Other states included in the navigation error model

The estimated errors are then used to correct the corresponding INS states or sensor-error estimates.

❑ **For example:**

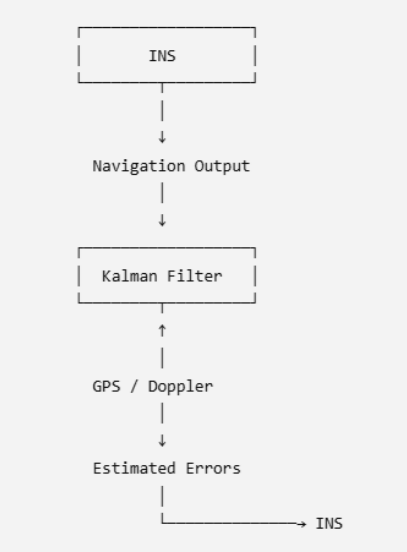
Estimated gyro bias → Correction → INS gyro-related navigation calculations

✓ This allows the INS to continue operating with a **reduced error**.

Closed-Loop Kalman Filter

4. Why is it called "Closed-Loop"?

It is called **closed-loop** because the correction information eventually returns to the system that generated the original navigation solution.
The complete loop is:



✓ So there is a **feedback loop**.

Closed-Loop Kalman Filter

5. Open-Loop vs. Closed-Loop

This distinction is extremely important.

➢ **Open-Loop**

- The Kalman filter estimates the errors and uses them to correct the **INS output**.
- INS → INS Output → Kalman Filter
- Corrected Output
- There is **no feedback into the INS**.

➢ **Closed-Loop**

- The Kalman filter estimates the errors and feeds the corrections **back into the INS**.
- INS→ INS Output → Kalman Filter → Corrections → INS

Therefore:

Open-loop → correct the output.

Closed-loop → correct the INS itself.

Closed-Loop Kalman Filter

6. Why can closed-loop be more robust?

- A major advantage of closed-loop mechanization is that the INS does not simply continue accumulating its original errors without internal correction.
- The Kalman filter continuously estimates errors and feeds appropriate corrections back into the navigation system.
- This can be particularly useful for **long-duration missions**, where small INS errors can accumulate significantly over time.
- Also notes that closed-loop operation can provide greater robustness when the **error model is simplified or imperfect**.
- However, this does **not** mean that closed-loop is always dramatically better.
- If the error model is sufficiently accurate, the performance of open-loop and closed-loop implementations can be **quite similar**.

7. A simple way to visualize it

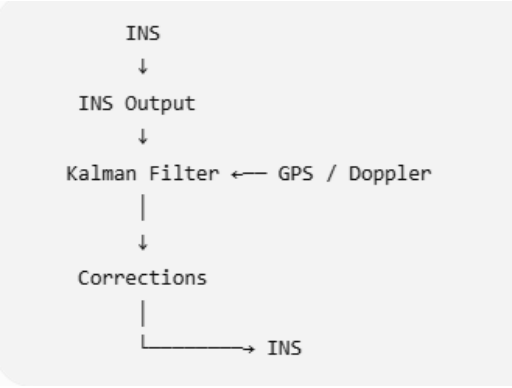
Think of the INS as a person measuring their position with a slightly inaccurate instrument.

❑ **Open-loop:**

The person continues using the instrument, and another system tells you: "Your reported position is off by 20 m."
You correct the **reported answer**, but the instrument itself remains unchanged.

❑ **Closed-loop:**

The external system says:
"Your instrument has a 20 m error.“ You use this information to **correct the navigation system itself**.
Then the INS produces a new, improved navigation solution.



A closed-loop Kalman filter mechanization is a configuration in which the Kalman filter estimates the INS errors and feeds the estimated corrections back into the INS, thereby correcting the INS navigation states or error sources rather than only correcting its output.

The feedback path is what makes it closed-loop.

Open-Loop Kalman Filter Mechanization

- **Open Loop vs Closed Loop**

Open Loop	Closed Loop
Corrects outputs	Corrects INS internally
INS remains uncorrected internally	INS maintains corrected solution
Simpler conceptual structure	More robust
Useful when pure INS output is desired	Useful for long missions
Corrections external to INS	Feedback corrections

GPS–INS Mechanization

- The **GPS–INS integration** is one of the most important applications of multisensor navigation.
- The basic idea is: GPS + INS
- The two systems have complementary characteristics. Each one compensates for important weaknesses of the other.

- **Why combine GPS and INS?**

INS — Inertial Navigation System

INS has several major advantages.

- **Short-term accuracy :** INS can provide very accurate navigation over short periods because it does not need an external reference during normal operation.
- **High bandwidth :** The INS operates at a high update rate, so it can respond quickly to aircraft motion.
- **Attitude information :** INS provides continuous estimates of: Roll , Pitch , Yaw/heading

This makes it particularly useful for aircraft navigation and flight-control applications.

- **Continuous navigation :** As long as the inertial sensors are operating, the INS can continuously calculate: Position , Velocity , Attitude

✓ However, the major problem is: INS error drifts with time
✓ Small errors in accelerometers and gyroscopes are integrated over time.
✓ So an INS is excellent in the **short term**, but its errors can grow over long periods.

GPS–INS Mechanization

GPS

GPS provides a different type of information.
Absolute position : GPS can provide an absolute position reference.

- **For example:** $P_{GPS} = 1000\,m$

This allows the navigation system to compare the GPS position with the INS position.
Velocity : GPS can also provide velocity information.
Long-term stability : Unlike a standalone INS, GPS does not suffer from the same type of continuously accumulating inertial drift.
Therefore, GPS can be used to **bound the long-term errors of the INS**.
However, GPS has an important limitation: GPS depends on an external signal
GPS measurements can be affected by:

- Signal blockage
- Interference
- Multipath
- Satellite geometry
- Temporary loss of signal

So GPS and INS have complementary strengths.

➢ The basic idea of GPS–INS integration

The combination can be summarized as:

INS >> Continuous navigation >> Position / Velocity / Attitude >> Kalman Filter >> GPS >> Position / Velocity

The **INS provides continuous, high-rate navigation**, while GPS provides an external reference that can be used to estimate and correct INS errors.

GPS–INS Mechanization

❑ **Example : Position measurement**

Suppose the INS estimates:

$$P_{INS} = 1002\ m$$

while GPS measures:

$$P_{GPS} = 1000\ m$$

The difference between the two measurements is called a **residual** or **innovation**, depending on the exact formulation.

We can write:

$$\boxed{r = P_{GPS} - P_{INS}}$$

Therefore:

$$r = 1000 - 1002$$

$$\boxed{r = -2\ m}$$

This tells us that there is a **2 m discrepancy** between the GPS and INS position estimates.

But the Kalman filter does not simply say: "The position is wrong by 2 m."

It tries to determine **why the discrepancy exists**.

Estimating the error states

This is one of the most important concepts in GPS–INS integration.
A measurement residual can contain information about several underlying errors.
For example, the Kalman filter may estimate:

- **Position error** : δP
- **Velocity error** : δV
- **Attitude error** : $\delta \theta$
- **Gyroscope bias** : b_g
- **Accelerometer bias** : b_a

and potentially other states depending on the system's error-state model.
So the filter is not simply performing: GPS Position → Position Correction
Instead, it is trying to determine the **combination of navigation and sensor errors that best explains the observed measurements**.

Linearizing a Nonlinear Range Measurement

A GPS range measurement is **inherently nonlinear**.
For example, the geometric range between the receiver and a satellite can be written as:

$$\rho = \sqrt{(x - x_s)^2 + (y - y_s)^2 + (z - z_s)^2}$$

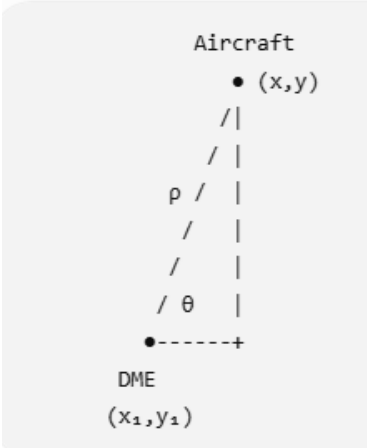
where:

- x, y, z= receiver position coordinates
- x_s, y_s, z_s= satellite position coordinates
- ρ= geometric range (distance from aircraft to DME)

The equation is nonlinear because the position variables appear inside a **square root and squared terms**.

This is simply the distance between:

- aircraft position: (x, y)
- DME station position: (x1,y1)



Linearizing a Nonlinear Range Measurement

➢ **Why do we need linearization?**

The basic Kalman filter is based on a **linear measurement model**: $\boxed{z = Hx + v}$

where:

- z= measurement
- x= state vector
- H= measurement matrix
- v= measurement noise

But our GPS range equation is nonlinear: $\rho = h(x)$

➢ **So we have a problem:**

Actual range equation $\rightarrow$ Nonlinear

but

Kalman filter $\rightarrow$ Linear model

Therefore, the book needs to **linearize** the range equation.

✓ Therefore, we cannot directly use the ordinary linear Kalman filter with the original range equation.
✓ We first need to approximate the nonlinear equation by a **linear equation**.
✓ This process is called: **Linearization**

Linearizing a Nonlinear Range Measurement

- **Actual/measured range** $\boxed{\rho}$

This comes from the DME measurement.

- **Predicted range** $\boxed{\rho^*}$

This is calculated using the **assumed/reference aircraft position** (x^*, y^*).
So:

$\rho =$ range based on actual position

while:

$\rho^* =$ range predicted from estimated position

Therefore their difference is:

$$\boxed{\boldsymbol{\Delta\rho = \rho^* - \rho}}$$

This is called the **range residual** or **measurement difference**.
Substitute into the Taylor expansion

$$\Delta\rho \approx \frac{\partial\rho}{\partial x}\Delta x + \frac{\partial\rho}{\partial y}\Delta y$$

Substitute:

$$\frac{\partial\rho}{\partial x} = \cos\theta \qquad \text{and} \qquad \frac{\partial\rho}{\partial y} = \sin\theta$$

giving:

$$\boxed{\Delta\rho = \Delta x\cos\theta + \Delta y\sin\theta}$$

$$\boxed{\Delta\rho = \rho^* - \rho = \Delta x\cos(\theta) + \Delta y\sin(\theta)}$$

Linearizing a Nonlinear Range Measurement

1. Linearization around an estimated position

The basic idea is to choose an **operating point**, usually the current estimated receiver position.
Suppose the estimated receiver position is: $\hat{x}, \hat{y}, \hat{z}$
The actual position can be represented as:

$$\boldsymbol{x = \hat{x} + \Delta x}$$
$$\boldsymbol{y = \hat{y} + \Delta y}$$
$$\boldsymbol{z = \hat{z} + \Delta z}$$

where: $\Delta x, \Delta y, \Delta z$ represent the **position errors**.
We then approximate the nonlinear range function around the estimated position using a **first-order Taylor expansion**.
Conceptually: Nonlinear GPS measurement → Choose operating point → Linearization → Linear measurement model → Kalman Filter

2. What does the linearized equation look like?

After linearization, the range measurement can approximately be expressed as:

$$\boxed{\Delta\rho \approx \frac{\partial\rho}{\partial x}\Delta x + \frac{\partial\rho}{\partial y}\Delta y + \frac{\partial\rho}{\partial z}\Delta z}$$

The partial derivatives describe how sensitive the measured range is to changes in the receiver position.
For example:

$$\frac{\partial\rho}{\partial x} = \frac{x - x_s}{\rho}$$

Similarly:

$$\frac{\partial\rho}{\partial y} = \frac{y - y_s}{\rho}$$
$$\frac{\partial\rho}{\partial z} = \frac{z - z_s}{\rho}$$

Therefore:

$$\boxed{\Delta\rho \approx \begin{bmatrix} \frac{x-x_s}{\rho} & \frac{y-y_s}{\rho} & \frac{z-z_s}{\rho} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix}}$$

This gives us the linear measurement relationship required by the Kalman filter.

Linearizing a Nonlinear Range Measurement

3. Why is this useful in GPS–INS?

In GPS–INS integration, we are usually interested in the **errors in the INS solution**, rather than directly estimating the entire absolute position from scratch.

For example:

$$\Delta P = \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix}$$

The GPS measurement provides information about these errors.

After linearization, the relationship can be represented in the general Kalman-filter form:

$$\boxed{z = Hx + v}$$

where the matrix $\boldsymbol{H}$ tells the filter how each measurement relates to the different error states.

GPS Clock Error Model

GPS is not only a **position problem**.
There is another important source of error: Receiver Clock Error
The GPS receiver has its own clock, and this clock is not perfectly synchronized with the GPS system time.
Therefore, two important clock-error states are considered:

1. Clock bias

The difference between the receiver clock and the reference GPS time. Clock Bias

2. Clock drift

The rate at which the receiver clock bias changes with time. Clock Drift

So the filter can contain: $\begin{bmatrix}\text{Clock Bias}\\ \text{Clock Drift}\end{bmatrix}$

- **Why does clock error matter?**

GPS measures **signal travel time**.
Conceptually: Range $= c \times$ signal travel time where $c$is the speed of light.
Therefore, even a small timing error can produce a significant apparent range error.
For example:
Receiver clock error → Timing error → Range measurement error → Position error

So the Kalman filter needs to account for receiver clock errors when processing GPS measurements.

- The two-state clock model therefore estimates: Clock Bias + Clock Drift

State GPS–INS Error Model

This is an important concept for understanding the simplified GPS–INS model.
The filter does not necessarily need to estimate every possible error in the navigation system.
Instead, the book introduces an **11-state error model** containing the most important states for the simplified GPS–INS system. Conceptually, the states can be grouped as follows.

1. Position Errors — 3 states

$$\boxed{\Delta x, \Delta y, \Delta z}$$

These represent errors in the estimated position.
2. Velocity Errors — 3 states

$$\boxed{\Delta V_x, \Delta V_y, \Delta V_z}$$

These represent errors in the INS velocity estimate.
3. Platform Misalignment / Attitude Errors — 3 states
The INS platform can have small angular errors.
These can be represented as:

$$\boxed{\phi_x, \phi_y, \phi_z}$$

where the three states represent small attitude or platform-misalignment errors.

- ❑ They can be interpreted conceptually as:
- Roll-axis misalignment
- Pitch-axis misalignment
- Azimuth/heading misalignment

These errors are important because an attitude error can cause the INS to resolve acceleration incorrectly.
For example: Attitude Error → Incorrect acceleration projection → Velocity Error → Position Error

State GPS–INS Error Model

4. GPS Receiver Clock Errors — 2 states
Finally: Clock Bias and Clock Drift
Therefore: $3 \text{ Position} + 3 \text{ Velocity} + 3 \text{ Attitude} + 2 \text{ Clock} = 11 \text{ States}$
The conceptual state vector is therefore: $x = [\Delta x \quad \Delta y \quad \Delta z \quad \Delta V_x \quad \Delta V_y \quad \Delta V_z \quad \phi_x \quad \phi_y \quad \phi_z \quad b_c \quad d_c]^T$
where $b_c$represents clock bias and $d_c$represents clock drift.

- **Why Error-State Instead of Full-State?**

This is a **very important concept** in GPS–INS integration.
Instead of asking the Kalman filter to estimate the complete absolute navigation state:
- Absolute Position
- Absolute Velocity
- Absolute Attitude

we make the filter estimate the **errors in the INS solution**.
For example: ΔP , ΔV , $\Delta \theta$
The Kalman filter therefore answers: **"How wrong is the INS?"** rather than: **"Where is the aircraft from scratch?"**

Example
Suppose: $P_{INS} = 1002\, m$, and $P_{GPS} = 1000\, m$,
The discrepancy is: $r = P_{GPS} - P_{INS}$
$r = -2\, m$
The Kalman filter uses this information together with its dynamic model to estimate states such as: $\Delta P, \Delta V, \Delta \theta$ and possibly other states.
Then the estimated errors can be used to correct the INS.
This is the basic idea behind an: Error-State Kalman Filter

Why Not Always Use Only 11 States?

The 11-state model is a **simplified error model**.
Some applications needs to more accuracy.
Real INS systems can have many additional error sources.

❑ **For example, we may add:**

Accelerometer biases

$$b_{ax}, b_{ay}, b_{az}$$

Gyroscope biases

$$b_{gx}, b_{gy}, b_{gz}$$

We could also include:

- Accelerometer scale-factor errors
- Gyroscope scale-factor errors
- Sensor misalignment errors
- Gravity disturbances
- Other instrument errors

The advantage is that a more detailed model can represent more of the real system's error behavior.
However, adding states increases the size of the Kalman filter.

Practical Considerations

This section moves from the **theoretical Kalman filter model** to the **practical implementation** of a GPS–INS navigation system.
There are four important issues:

1. Measurement Synchronization
2. Measurement Editing
3. Filter Tuning
4. Sequential Measurement Processing

1. Measurement Synchronization

The GPS and INS measurements must correspond to the **same time**.
Ideally, we want: $GPS(t)$, to be compared with: $INS(t)$
at the same instant.

❑ **Example**

Suppose the GPS measurement is obtained at: $t = 10.000\ s$, but the INS prediction being used is from $t = 10.100\ s$
There is a: $\boxed{0.100\ s}$ time mismatch.
The aircraft may have moved significantly during this interval.
Therefore, when we calculate the residual: $r = z_{GPS} - z_{INS}$
the residual may appear larger than expected.
But the problem may **not actually be an INS navigation error**.
It could simply be: Time synchronization error

Practical Considerations

2. Measurement Editing

The Kalman filter should **not automatically accept every measurement**.
Sometimes a measurement can be:

- corrupted,
- inconsistent,
- affected by interference,
- affected by multipath,
- or simply outside the expected range.

The filter can check whether the measurement is reasonable before using it.

Residual / Innovation Test

Suppose the measurement residual is: $r = z - H\hat{x}$
The filter can compare the magnitude of the residual with its **expected uncertainty**.
If the residual is reasonably small: $|\ r\ | \approx$ expected , the measurement can be accepted.
But if: $|\ r\ | \gg$ expected , then the measurement may be considered suspicious.
This process is called: Measurement Editing or Reasonableness Testing

❑ Example

Suppose the expected measurement uncertainty is approximately: $\sigma = 2\,m$, and the observed residual is $r = 1.5\,m$, This may be perfectly reasonable.
But suppose: $r = 50\,m$, A residual of 50 m is much larger than what the filter expected.
The system may therefore reject or edit that measurement rather than allowing it to strongly influence the navigation solution.

➢ Why is this important?

Imagine that the GPS suddenly produces a bad measurement.
If the Kalman filter accepts it without checking:

Bad GPS measurement → Kalman Filter → Incorrect error estimate → Incorrect correction → Navigation solution degraded

Therefore: Measurement Editing protects the filter from bad measurements.

Practical Considerations

3. Filter Tuning

Another major practical issue is choosing appropriate values for the covariance matrices: $\boxed{Q}$ and $\boxed{R}$, These values strongly affect how the Kalman filter behaves.

A. Process Noise Q

Qrepresents uncertainty in the **system model/process**.

In simple terms: How much do we trust our model of how the system evolves?

❑ **If $Q$is too small >>** The filter assumes that the model is very reliable. $Q \downarrow$

Therefore, it tends to rely heavily on the prediction.

Conceptually:

Q too small >> Strong confidence in model >> Less willingness to change prediction

If the real system behaves differently from the model, this can cause problems.

❑ **If $Q$is too large**

The filter assumes that the model is relatively uncertain. $Q \uparrow$

Therefore, the filter becomes more willing to adjust its estimate based on measurements.

Q too large >> Less confidence in model >> More reliance on measurements

Practical Considerations

B. Measurement Noise R

R represents the uncertainty of the **measurement**.

In simple terms: How much do we trust the measurement?

- **Small R**

$R \downarrow$ means: Trust the measurement more

For example, if GPS is believed to be highly accurate, its measurement noise covariance may be relatively small.

- **Large R**

$R \uparrow$ means: Trust the measurement less

The filter will rely relatively more on its prediction.

A useful way to remember Q and R

Q , Think: **"How much do I trust my model?"**

R , Think: **"How much do I trust my measurement?"**

So:

$Q \rightarrow$ Model uncertainty

$R \rightarrow$ Measurement uncertainty

Practical Considerations

4. Sequential Measurement Processing

In a practical navigation system, there may be many measurements available at approximately the same time.

For example, GPS may provide measurements from several satellites:

- Satellite 1 → z_1
- Satellite 2 → z_2
- Satellite 3 → z_3
- Satellite 4 → z_4

One approach would be to combine all of them into one large measurement vector:

$$z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix}$$

However, it is also possible to process them **sequentially**.

Practical Considerations

➢ **Sequential approach**

Instead of processing all measurements simultaneously:

z_1 , z_2 , z_3 , z_4 >> One large measurement update

we can process them one at a time:

Prediction

↓

Process z_1

↓

Updated estimate

↓

Process z_2

↓

Updated estimate

↓

Process z_3

↓

Updated estimate

↓

Process z_4

↓

Final updated estimate

✓ This is called: Sequential Measurement Processing or **scalar measurement processing** when individual scalar measurements are processed separately.

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